

Injury to the Anterior Cruciate Ligament During Alpine Skiing

A Biomechanical Analysis of Tibial Torque and Knee Flexion Angle

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Background: The anterior cruciate ligament has been shown to be particularly susceptible to injury during alpine skiing. Tibial torque is an important injury mechanism, especially when applied to a fully extended or fully flexed knee.

Purpose: We wanted to record the forces generated in the anterior cruciate ligament with application of tibial torque to cadaveric knees in different positions.

Study Design: Controlled laboratory study.

Methods: Thirty-seven fresh-frozen cadaveric knees were instrumented with a tibial load cell that measured resultant force in the anterior cruciate ligament while internal and external tibial torques were applied to the tibia at full extension, 90° of flexion, full flexion, and forced hyperflexion.

Results: At each knee flexion position, mean force generated by 10 N·m of internal tibial torque was significantly higher than the mean generated by 10 N·m of external tibial torque. Mean forces generated by tibial torque at 90° of flexion were relatively low. During flexion-extension without tibial torque applied mean forces were highest (193 N) when the knee was hyperflexed.

Conclusions: Application of internal tibial torque to a fully extended or fully flexed knee represents the most dangerous loading condition for injury from twisting falls during skiing.

Clinical Relevance: Understanding of the mechanisms of falls can be used to design better equipment and to better prevent or treat injury.

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Lower extremity injuries in alpine skiing have significantly decreased over the last several decades.^{1,6,7} In particular, tibia and ankle fractures have decreased by 90%.⁷ Despite this trend and the advancements in the design of ski bindings and boot release systems, injuries to the ACL among alpine skiers have increased.^{1,6,7} Currently, 25% to 30% of all ski-related knee injuries involve the ACL.¹³

The mechanisms of injury particular to alpine skiing must be understood so that clinicians can appropriately assess, treat, and prevent ACL injury in the alpine skier. Numerous injury mechanisms have been proposed, but

most are based on skier recall and videotape analysis of falls in which ACL tears have occurred. Many skiers have difficulty remembering the position of their limbs when they fell. Only a limited number of skier accidents can be studied from videotapes, and limb alignment must be assessed through the thick protective clothing worn by the skier.

Although most injury mechanisms that cause ACL injury involve twisting, in alpine skiing two proposed injury mechanisms are associated with a hard landing from a jump. McConkey¹⁰ reported on 15 expert skiers whose ACL tears resulted from the contraction of the quadriceps muscle to recover from an out-of-control sitting-back posture or to gain control after landing from a jump. Five years later, Geyer and Wirth³ observed isolated ACL ruptures in downhill racers who, to prevent a backward fall after landing from a jump, elicited a massive quadriceps contracture that displaced the tibia anteriorly and rup-

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tured the ACL. This mechanism of injury appears to occur only in elite or expert skiers whose quadriceps muscles can generate the large forces necessary to displace the tibia anteriorly. Ettlinger et al.¹ proposed an additional mechanism of injury that also involves landing from a jump. In this scenario, the skier lands on one leg in an off-balance condition, and the force of impact drives the ski boot anteriorly with respect to the femur, rupturing the ACL. This is termed the "boot-induced" mechanism of ACL injury.

The remainder of the ACL injury mechanisms in alpine skiing proposed in the literature involve twisting of the knee. Internal and external tibial torque applied to the knee at various flexion angles have been proposed as mechanisms of injury.^{1,2,4,5,9,11,13} Once again, these mechanisms of injury are based on descriptions from skiers or from analysis of videotapes. Jarvinen et al.⁵ studied 51 patients who sustained an ACL rupture during downhill or cross-country skiing; 47% of the skiers reported a valgus-external rotation mechanism and 41% reported a flexion-internal rotation mechanism. Two additional skiers described a hyperextension-internal rotation mechanism. Ettlinger et al.¹ described the "phantom foot" mechanism of ACL injury, in which an off-balance skier falls backward, hyperflexing the knee with all his or her weight on the inside edge of the tail of the downhill ski. This movement causes the ski to turn uphill, which rotates the tibia internally relative to the femur. Although this sequence of events has been observed repeatedly in videotapes of skier falls, there has been no experimental verification of this proposed mechanism of ACL injury. The purpose of this study was to directly record forces generated in the ACL during application of tibial torque to cadaveric knee specimens at four knee flexion angles representing four mechanisms of ACL injury during skiing.

MATERIALS AND METHODS

Thirty-seven fresh-frozen knee specimens were used for this study; the mean age of the donors was 57.1 years (range, 38 to 84). Each specimen was manually tested for stability and inspected visually for intraarticular pathologic conditions, such as arthritis or meniscal damage, through a small medial parapatellar arthrotomy incision. The tibia and femur were each sectioned at midshaft and cleaned of soft tissue to within 10 cm of the joint line. The ends of the bones were potted in cylindrical molds of polymethyl methacrylate for gripping in the testing fixtures. A bone cap containing the tibial insertion of the ACL (centered at the tibial footprint) was mechanically isolated by using a coring reamer. It was then attached to a tibial load cell that measured resultant force in the ligament. An electronic torque handle was fastened to the end of a tibial extension shaft connected to the potted tibia. Applied tibial torque and resultant ACL force were recorded simultaneously on a personal computer using an A/D interface board and Lab Tech Notebook software package (LabTech, Andover, Massachusetts). Details of specimen preparation and the ACL bone cap isolation technique can be found in a prior publication.⁸

Testing was performed with the knee at 0° of flexion (as defined by application of a 2-N·m extension moment), 90° of flexion, full manual flexion, and at forced hyperflexion (as determined by hanging an appropriate weight to a roller bearing on the tibial extension shaft to produce a 15-N·m flexion moment). Application of the weight to the roller bearing on the tibial extension shaft did not alter the magnitude of the applied tibial torque. For each flexion position, the tibia was first placed in neutral rotation. Then tibial torque was applied manually through the instrumented handle at a loading rate of approximately 10 deg/sec (first internally and then externally) to a level of 10 N·m.

A two-way repeated measures analysis of variance was used to determine the significance of differences between mean values of ACL resultant force between test conditions. The test conditions were level of applied tibial torque (0 N·m, 10 N·m external, and 10 N·m internal torque) and position of knee flexion (0°, 90°, full flexion, 15 N·m hyperextension moment). The level of significance was $P < 0.05$.

RESULTS

For all four knee flexion positions tested, the mean ACL forces generated by 10 N·m of internal torque were significantly greater than the corresponding means for 0 N of applied tibial torque and 10 N·m of external tibial torque ($P < 0.05$) (Figs. 1 through 4).

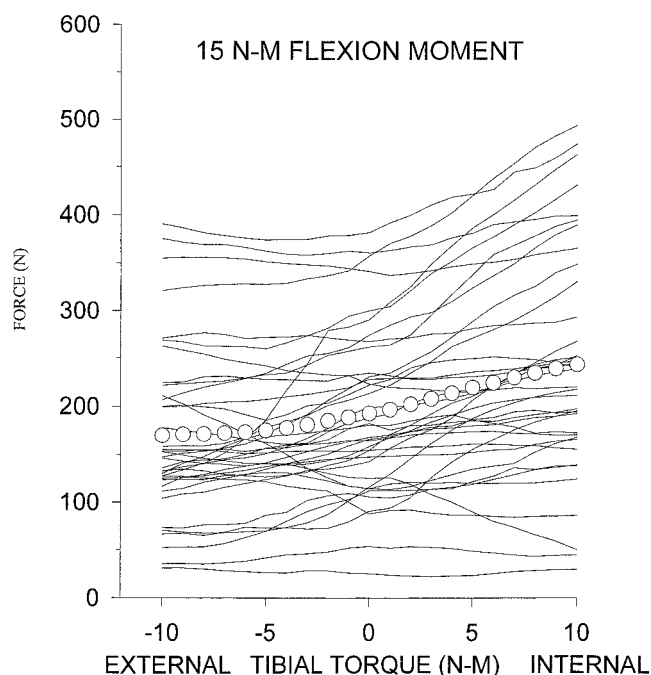


Figure 1. Test curves of ACL resultant force versus applied tibial torque for all knee specimens with 15 N·m of hyperflexion moment. Mean values of ACL force are shown by white symbols.

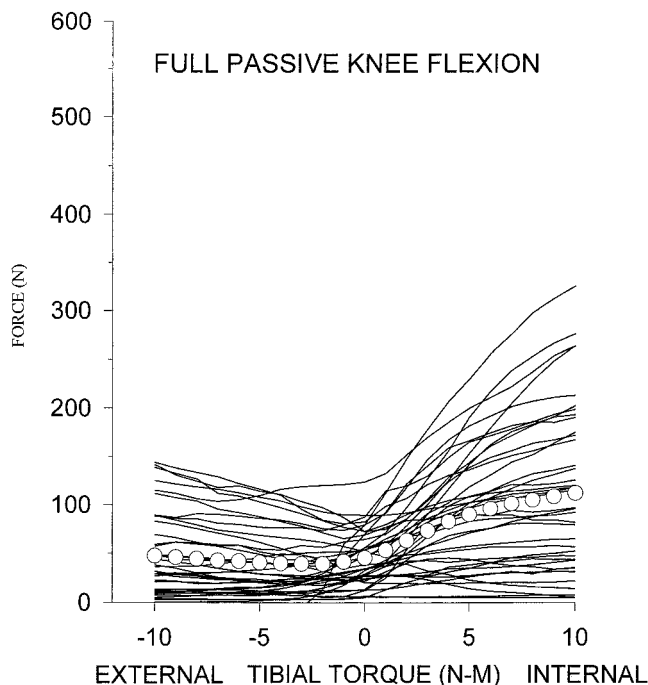


Figure 2. Test curves of ACL resultant force versus applied tibial torque for all knee specimens at full manual flexion. Mean values of ACL force are shown by white symbols.

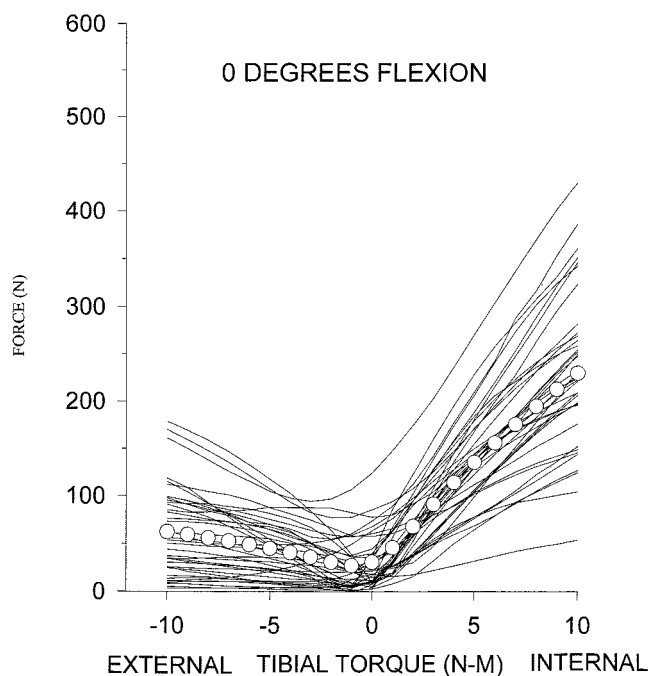


Figure 4. Test curves of ACL resultant force versus applied tibial torque for all knee specimens at 0° of flexion. Mean values of ACL force are shown by white symbols.

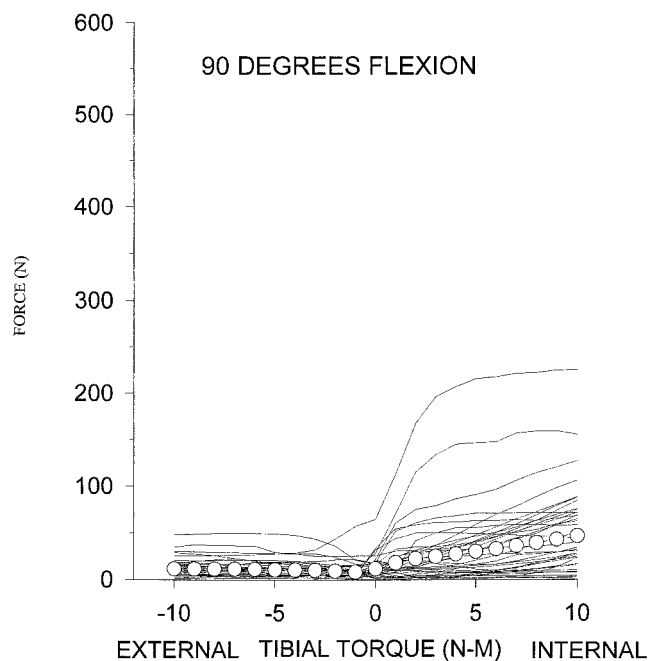


Figure 3. Test curves of ACL resultant force versus applied tibial torque for all knee specimens at 90° of flexion. Mean values of ACL force are shown by white symbols.

During knee flexion-extension without the application of tibial torque, the mean ACL force was highest (193 N) with hyperflexion of the knee (Fig. 1). With the addition of

internal tibial torque, the mean ACL force with the knee in hyperflexion (243 N) was significantly greater than that at full flexion (112 N) and 90° of flexion (47 N). It was not significantly different from the corresponding mean for internal torque applied at full extension (230 N) (Figs. 1 through 4).

For external tibial torque, mean ACL force with the knee in hyperflexion (170 N) was significantly greater than the corresponding means for full flexion (48 N), 90° of flexion (12 N), and full extension (62 N) (Figs. 1 through 4).

DISCUSSION

The number of alpine skiing injuries have declined over the last 2 decades, most likely because of advances in binding and ski boot design. However, knee injuries, particularly ACL tears, have been on the rise. Prevention of knee injuries will depend on a better understanding of the mechanisms of injury and the positions of the knee that put the ACL at risk. To date, most reports of mechanisms of injury are based on skier recall and videotape analysis. Biomechanical analysis of mechanisms of injury to the ACL is, therefore, important to confirm or dispute currently proposed mechanisms.

In this study, biomechanical analysis of ACL forces was conducted under quasistatic conditions with no active knee musculature. Therefore, forces generated by tibial torque during a fall could be considerably higher than those recorded in this study. Applied tibial torque levels used in this study were limited to 10 N·m by the strength of the ACL-bone cap construct. Application of more than

10 N·m of tibial torque disrupted the ligament's attachment to the load cell. In addition, excessive ACL forces and the possibility of capsular damage to the knees were also reasons for limiting the tibial torque to 10 N·m, particularly with the knee in the hyperextended position. The highest ACL force recorded in an individual specimen (approximately 500 N) represented approximately one-quarter of the ultimate strength of this ligament (2160 N) in a young person.¹⁴ With greater tibial torque, ligament forces would most likely continue to increase.

When muscular contraction and its effects on tibial torque were analyzed in a previous study, we found that mean tibial torques generated by maximal contraction of the knee musculature ranged from 30 to 70 N·m, depending on the test condition and knee flexion angle.¹² If these muscles are recruited during a fall, they are capable of resisting applied tibial torque, thereby reducing force generated in the ACL. However, if the knee is in an awkward position during a fall (such as hyperflexion with internal tibial rotation) or the fall is too rapid for the skier to react, the knee musculature may not be activated to produce significant resistive tibial torque. In these instances, high levels of torque may be imparted directly to the knee joint, which in turn could generate dangerous levels of force in the ACL.

Our data support some of the mechanisms of injury to the ACL described in the literature. In particular, our data support the "phantom foot" ACL injury mechanism proposed by Ettlinger et al.,¹ which involves internal tibial rotation of a flexed knee. Although we found that, on average, addition of internal tibial torque to a fully flexed or hyperflexed knee further increased ACL force, many knee specimens demonstrated little or no change in ACL force with either internal or external tibial torque (Figs. 1 and 2). Thus, our data suggest that hyperflexion may be the predominant factor for the ACL injury mechanism described by Ettlinger et al. and not the twisting component of the fall. Hyperflexion may also contribute to ACL injury in the cases of elite skiers who rupture their ACL attempting to recover from a jump. In contrast, ACL forces generated at full extension were clearly increased by addition of internal tibial torque for all specimens in the study (Fig. 4). This finding corroborates the common skiing injury mechanism of internal tibial rotation of a fully

extended or hyperextended knee, which occurs when a skier crosses the ski tips while falling forward.

In summary, given the increased graft forces they produce, hyperflexion and hyperextension of the knee with the addition of internal tibial torque appear to be mechanisms of injury for the ACL in alpine skiers. Continued advancements in the design of skiing equipment and protective gear as well as skier education are necessary if there is to be success in reducing the number of knee injuries that occur on the slopes.

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